SOLVING DYNAMIC MODELS WITH RATIONAL INATTENTION

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Abstract

We provide a new algorithm to solve dynamic stochastic general equilibrium models with rational inattention (RI-DSGE). Existing procedures update the equilibrium and attention jointly, through *tâtonnement* on the aggregate paths. We decouple the two: an inner loop solves the general equilibrium given attention, and an outer loop updates attention given the equilibrium. The inner loop relies on an exact mapping of attention into expectation matrices, under which the equilibrium is a single linear system in the sequence space. Our algorithm thus converges in few iterations, which allows the attention problems to be solved on an almost exact representation of the equilibrium dynamics. Applied to models from the literature, our algorithm recovers the published equilibria up to 1,500 times faster, with the largest gains where existing procedures must take small steps to remain stable.

Keywords: rational inattention, information frictions, expectations, DSGE, computational methods.

JEL Codes: C61, C63, D83, D84, E32, E52

[†] The toolbox implementing the method will be made public in the future; in the meantime, it is available upon request.

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1. Introduction

Dynamic stochastic general equilibrium (DSGE) models are one of the main macroeconomic frameworks for quantitative research and policy analysis. Considerable effort has gone into making solution methods for such models accessible, to the point that the computational burden is often no longer a concern (see Adjemian et al. 2011 and Auclert et al. 2021 for widely used toolboxes). Most of these methods can be modified to accommodate departures from perfect information rational expectations (Mankiw and Reis, 2007; Gabaix, 2020; Angeletos and Huo, 2021), but only with the degree of information frictions taken as given; limited attention enters the model like any other exogenous source of sluggish adjustment.

Sims (1998)'s *rational inattention* (RI) instead derives the degree of information frictions from a microfounded problem: limited attention is an equilibrium outcome that responds to policy and the economic environment (see Maćkowiak and Wiederholt, 2015, 2025 and Afrouzi and Yang 2021). RI pairs particularly well with linearized DSGE models, in which optimal attention takes the form of Gaussian signals about the state of the economy (Maćkowiak, Matějka, and Wiederholt, 2018). In expectation, actions satisfy the optimality conditions of the perfect information economy, and the Kalman filter delivers them easily. The drawback is that solving RI-DSGE models takes considerable know-how: it requires finding "the fixed point of an economy in which the optimal signal of an agent depends on the signals chosen by other agents" (Maćkowiak and Wiederholt 2025, p. 274), for which no *one-shot* solution method exists.

In this paper, we propose a *new* algorithm for solving RI-DSGE models that is substantially faster than existing procedures and can thus be made arbitrarily accurate. Existing procedures revise a guess of the aggregate paths jointly with attention, a form of *tâtonnement*, so general equilibrium feedbacks—such as strategic complementarities or interactions between agents' attention choices—are resolved only across iterations.[1] Our algorithm separates the two. Given attention, an inner loop solves exactly for the paths of all endogenous variables in the sequence space. The outer loop then updates attention given the equilibrium paths.

How our method departs from existing procedures is best understood by analogy with value function iteration and policy iteration (Howard, 1960), two methods for solving Bellman equations. Value function iteration applies the Bellman operator and simultaneously updates the value and the policy at every step, so both converge together, often slowly. Policy iteration instead evaluates the value of the

[1] In practice, the aggregate paths are represented by finite-order ARMA processes or truncated moving averages, refitted at every iteration.

Table 1: Solution Time

		Time (seconds)		
Model	Shock	Published	Ours	Speed-up
AY21	Monetary & technology	219	85.0	2.6
MW15	Monetary	925	5.1	181
	TFP	125	4.5	28
MW25	Technology, $h = 0$	14,743	9.6	1,536
	Technology, $h = 2$	3,674	39.8	92

Notes: In MW25, the technology shock is anticipated h periods ahead ($h = 0$: no news). We cannot attest whether the published damping steps are optimized for speed; for ours, we report the fastest among a few candidate steps. Hardware: AMD EPYC 7532, 4 cores; Julia 1.10.

current policy exactly before improving it, and typically converges in far fewer iterations. Current RI-DSGE solvers resemble value function iteration: each iteration takes aggregate paths as given and updates both attention and the equilibrium. Our algorithm resembles policy iteration: the inner loop solves the equilibrium implied by the current attention exactly before the outer loop updates attention. Like existing procedures, it is guess-and-verify—but over attention only—with each candidate checked against the exact equilibrium it implies.

Solving the inner loop in the sequence space provides a straightforward way to resolve the general equilibrium exactly at every iteration. Given attention, the equilibrium is a single linear system in the sequences, which delivers the paths of all endogenous variables at once with the boundary conditions imposed. Existing solvers have no such step: they obtain the paths that involve forward-looking variables—in the targets of the attention problems or in the rest of the equilibrium—by iterating sums forward or in separate partial equilibrium problems. We show how to map attention into expectation matrices, extending the treatment of sticky expectations in Auclert, Rognlie, and Straub (2020) to rational inattention, where, given the solution of the attention problems, the mapping follows from filtering on the history of innovations. Once expressed this way, RI-DSGE models also benefit from the tools of the sequence space, such as the representation of models as directed acyclic graphs of blocks (Auclert et al., 2021).

Decoupling attention from the general equilibrium also makes our method more robust. When general equilibrium feedbacks are strong, existing procedures remain stable only with very small steps. Over the many iterations then required, even small

errors in solving the attention problems or the rest of the equilibrium can accumulate into spurious dynamics. Compact representations of the paths, often used to keep so many iterations affordable, compound the problem.[2] Because our inner loop resolves the general equilibrium exactly, the contraction rate of our outer loop depends only on how strongly the paths move attention, which is typically weak (Section 5), so larger steps can be taken and fewer iterations suffice. The attention problems can then be solved on the almost exact MA(T) representation, where T is the truncation horizon of the sequences.

We demonstrate the effectiveness of our method by solving RI-DSGE models from the literature.[3] Table 1 reports the solution times: our algorithm reaches the same equilibrium 2.6 to 1,536 times faster than the published methods. The gains are largest where existing procedures must take small steps to remain stable, as in Maćkowiak and Wiederholt (2025). The increased speed renders likelihood-based estimation possible, even on a laptop, which we illustrate on the model of Maćkowiak and Wiederholt (2015). We provide a toolbox that implements our algorithm.

RI-DSGE models in the literature. Maćkowiak and Wiederholt (2009) embed inattention in the price-setting framework of Woodford (2003), in which nominal aggregate demand follows an exogenous process. Maćkowiak and Wiederholt (2015) build a small-scale DSGE model wherein households and firms are both inattentive. Afrouzi and Yang (2021) return to inattentive price setters, but with an explicit demand side, closed by an interest rate rule whose changes they study. Song and Stern (2025) split firms into attentive and inattentive types facing monetary policy shocks. Maćkowiak and Wiederholt (2025) build a real business cycle model in which inattentive firms and households face technology shocks that can be anticipated. All of these models are solved by iterating on the paths of aggregate variables.[4]

Within each of these models, the attention problems are solved either by numerical optimization or, more efficiently, by iterating on their first-order conditions (Afrouzi and Yang, 2021; Miao, Wu, and Young, 2022). We propose a way to accelerate these iterative procedures on large state spaces, which may be of independent interest.

Layout. The remainder of the paper is structured as follows. Section 2 represents the equilibrium, given the expectation matrices, as a single linear system in the sequence space.. Section 3 formulates the attention problem and shows how its

[2]Fitting persistent dynamics with low-order VARMA processes often produces complex roots, which introduce spurious oscillatory components.

[3]Not to be confused with models in which attention is chosen outside the general equilibrium, such as Pfäuti (2023).

[4]Afrouzi and Yang (2021) also work in the sequence space, but still through *tâtonnement*.

solution maps into the expectation matrices. Section 4 describes the algorithm. Section 5 applies it to RI-DSGE models from the literature. Section 6 concludes.

2. Equilibrium Representation

We consider economies whose equilibrium can be written as a system of linear stochastic difference equations.[5] Fluctuations arise from a vector of n_ε structural innovations $\boldsymbol{\varepsilon}_t$, Gaussian and i.i.d. over time. Collecting the n_x endogenous variables in $\boldsymbol{x}_t$, the equilibrium conditions read:

$$E_t\left[\Phi_+\boldsymbol{x}_{t+1} + \Phi_0\boldsymbol{x}_t + \Phi_-\boldsymbol{x}_{t-1} + \Phi_\varepsilon\boldsymbol{\varepsilon}_t\right] = \mathbf{0}, \tag{1}$$

and the solution of the model expresses the endogenous variables as a function of the state variables—their own lags and the innovations:

$$\boldsymbol{x}_t = \Pi_-\boldsymbol{x}_{t-1} + \Pi_\varepsilon\boldsymbol{\varepsilon}_t. \tag{2}$$

Several well-known methods solve such systems under perfect information and rational expectations (Blanchard and Kahn 1980, Klein 2000, Sims 2002). Under imperfect information, the state must also carry the agents' beliefs, and when agents learn from endogenous variables—so that the forecasts of others matter—the equilibrium may admit no finite-state representation such as Equation (2) (Townsend, 1983).[6]

To keep the state independent of the information structure, we work instead with the moving-average representation of the equilibrium presented in Section 2.1, whose state is the truncated history of innovations.[7] Information frictions then alter only the mapping from the sequences of the targets—the actions an agent would take under perfect information given the aggregate paths—to those of the actions, given by the expectation matrices of Section 2.2.

2.1. Sequence Space. By the Wold decomposition, any linear equilibrium can be expressed as a linear function of current and past innovations:

$$\boldsymbol{x}_t = \sum_{\tau=0}^{T-1} \psi_\tau^x\, \boldsymbol{\varepsilon}_{t-\tau}, \tag{3}$$

[5] Typically in log-deviations from the non-stochastic steady state; i.e., $x_t := \log X_t - \log X$.

[6] Nimark (2007) obtains a finite-state approximation by truncating the hierarchy of higher-order expectations included in the state.

[7] Lorenzoni (2009) similarly truncates a history, of aggregate variables rather than innovations, and filters on the truncated state.

where ψ^x_τ is the response of $\boldsymbol{x}_t$ to the innovations that occurred τ periods earlier. The truncation at horizon T, the only approximation of the representation, is chosen so that the responses have converged. We stack the entries of $\psi^x_0, \ldots, \psi^x_{T-1}$ into one vector, the *sequence* of x, denoted $\boldsymbol{x}$, which collects the impulse responses of x to every innovation at every horizon. Under perfect information rational expectations, $\psi^x_\tau = \Pi_-^\tau \Pi_\varepsilon$, and the sequence of a variable is also, by certainty equivalence, its perfect foresight path following an innovation at date 0 that no one anticipated—an "MIT" shock.

Since each equilibrium condition is linear, it holds separately for each innovation and each horizon, and the equilibrium can be written as a linear relation between the sequences of the variables it involves:

$$\boldsymbol{\Phi}_x \, \boldsymbol{x} + \boldsymbol{\Phi}_\varepsilon \, \boldsymbol{\varepsilon} = \boldsymbol{0}, \tag{4}$$

where $\boldsymbol{\varepsilon}$ is the sequence of the innovations themselves—a unit impulse at horizon 0 for each innovation—and $\boldsymbol{\Phi}_x$ and $\boldsymbol{\Phi}_\varepsilon$ stack Φ_-, Φ_0, Φ_+ and Φ_ε horizon by horizon and innovation by innovation. Equation (4) is the sequence-space representation of Auclert et al. (2021), a stacked system in the tradition of Anderson and Moore (1985).

2.2. Expectations. Some of the variables in $\boldsymbol{x}_t$ are chosen by inattentive agents, that is, agents who face information frictions. These agents belong to groups indexed by g, such as firms or households, and each group is populated by a continuum of agents $\iota \in [0, 1]$. Agents in group g share the same optimality conditions, which define the group's *target* $\boldsymbol{x}^*_{gt}$ (e.g., the profit-maximizing price for a firm, or the consumption implied by the Euler equation for a household), the action they would take under perfect information, given the aggregate paths. Each agent sets its action to its best estimate of the target given its information set $\mathcal{I}_{\iota gt}$:

$$\boldsymbol{x}_{\iota gt} = E\left[\boldsymbol{x}^*_{gt} \mid \mathcal{I}_{\iota gt}\right]. \tag{5}$$

Only the cross-sectional average of the actions within each group, $\bar{\boldsymbol{x}}_{gt} := \int_0^1 \boldsymbol{x}_{\iota gt} \, d\iota$ for group g, enters the vector of endogenous variables $\boldsymbol{x}_t$.

We focus on information sets consisting of Gaussian signals about the current and past state of the economy, with noise that is idiosyncratic across agents. Section 3 describes these signals and how agents choose them. Each agent's estimate of the target is then linear in the innovations and in its signal noise, and the noise integrates to zero over the continuum, so the sequence of the average action is a linear function of the target's sequence, given by a matrix $\boldsymbol{\mathcal{E}}_g$:

$$\bar{\boldsymbol{x}}_g = \boldsymbol{\mathcal{E}}_g \, \boldsymbol{x}^*_g. \tag{6}$$

We call $\boldsymbol{\mathcal{E}}_g$ the *expectation matrix* of group g, following Auclert, Rognlie, and Straub (2020), who introduce it in the context of sticky expectations. Consider first the case of a single structural shock, $n_\varepsilon = 1$, and an innovation at date 0. Row h of $\boldsymbol{\mathcal{E}}_g$ collects the average estimates held at date h: its entry in column s is the average estimate, at date h, of an innovation at date $h - s$. The diagonal entry of row h, in column $s = h$, is thus how much of the actual innovation agents have learned by date h, and the other entries of the row how much of it they attribute to innovations at other dates. Under perfect information the expectation matrix is the identity; if agents learn nothing it is zero; if they eventually learn the innovation, the diagonal entry of row h tends to one and the other entries of the row to zero as h grows. When the group chooses several actions, all of them are based on these same estimates, so the same matrix applies to each component of the target.

Expectation matrices also accommodate the confounding of structural shocks when $n_\varepsilon > 1$: a signal that does not separate the shocks (on their sum, for instance) leads agents to attribute part of one innovation to another. Since the sequences are stacked as vectors across innovations and horizons, the expectation matrix is then made of $n_\varepsilon \times n_\varepsilon$ blocks of size $T \times T$, one for each pair of innovations. Each block has the structure of the single-shock case: the block pairing an actual innovation with a second one collects the average estimates of the second following the first. Without confounding, the off-diagonal blocks are zero and each diagonal block is the expectation matrix of a single shock; with confounding, the off-diagonal blocks measure how much of each innovation is attributed to the others.

The sequence-space representation and the expectation matrices together make up the inner loop of our algorithm: given the expectation matrices, the equilibrium is a single linear system in the sequences, solved at once for the paths of all endogenous variables. The next section describes how, given these paths, the attention problems are solved and the expectation matrices obtained.

3. Attention

This section lays out the microfoundations of the expectation matrices, and with them the outer loop of our algorithm: given the equilibrium paths, each agent solves an attention problem that determines its signals about the state, and the Kalman filter maps these signals into the expectation matrix of the group. Since the attention problem has the same form for every group, we drop the group index g throughout.

3.1. Signals. The state of the economy is the history of innovations, which evolves by shifting one period and adding the new innovations:

$$\boldsymbol{\xi}_t := \begin{bmatrix} \boldsymbol{\varepsilon}_t \\ \vdots \\ \boldsymbol{\varepsilon}_{t-T+1} \end{bmatrix}, \qquad \boldsymbol{\xi}_{t+1} = \boldsymbol{F}\boldsymbol{\xi}_t + \begin{bmatrix} \boldsymbol{\varepsilon}_{t+1} \\ \mathbf{0} \end{bmatrix}. \tag{7}$$

Every variable, the target included, is then a linear function of $\boldsymbol{\xi}_t$, with its impulse responses Equation (3) as loadings:[8]

$$\boldsymbol{x}_t^* = \boldsymbol{\Psi}^*\boldsymbol{\xi}_t, \qquad \boldsymbol{\Psi}^* := \begin{bmatrix} \psi_0^{x^*} & \cdots & \psi_{T-1}^{x^*} \end{bmatrix}. \tag{8}$$

Agents receive noisy signals about the state:

$$\boldsymbol{S}_{\iota t} = \boldsymbol{\Gamma}\,\boldsymbol{\xi}_t + \boldsymbol{\nu}_{\iota t}, \qquad \boldsymbol{\nu}_{\iota t} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\nu}}), \tag{9}$$

where the noise $\boldsymbol{\nu}_{\iota t}$ is i.i.d., independent of the innovations, and idiosyncratic, so that it integrates to zero over the group. Signals of this form—a linear function of the current state plus idiosyncratic Gaussian noise—are optimal under the attention problem of Section 3.2 (see Maćkowiak, Matějka, and Wiederholt 2018). A signal on a variable—a price, output, the interest rate, last period's sales—is a signal on $\boldsymbol{\xi}_t$ with the impulse responses of that variable as loadings, which is why $\boldsymbol{\Gamma}$ depends on the equilibrium. The signals can be specified exogenously, in which case $\boldsymbol{\Gamma}$ and $\boldsymbol{\Sigma}_{\boldsymbol{\nu}}$ are given and the expectation matrix follows directly from Section 3.3, or chosen by the agents, as under rational inattention.

3.2. Dynamic LQG-RI Problem. Under rational inattention the signals are not given: attention is costly, and each agent designs its signals recognizing that processing more information reduces the expected losses from suboptimal actions but comes at a higher attention cost. We formulate the problem as a dynamic linear-quadratic-Gaussian rational inattention (LQG-RI) problem. In period $t = -1$, agent ι chooses an attention strategy—a pair of matrices $\{\boldsymbol{\Gamma}, \boldsymbol{\Sigma}_{\boldsymbol{\nu}}\}$ collecting the loadings and the noise of its signals—to solve:[9]

$$\max_{\{\boldsymbol{\Gamma}, \boldsymbol{\Sigma}_{\boldsymbol{\nu}}\}} \left\{ \sum_{t=0}^{\infty} \beta^t E_{\iota,-1} \left[\frac{1}{2} \left(\boldsymbol{x}_{\iota t} - \boldsymbol{x}_t^* \right)' \boldsymbol{\Theta} \left(\boldsymbol{x}_{\iota t} - \boldsymbol{x}_t^* \right) \right] - \lambda \sum_{t=0}^{\infty} \beta^t I(\boldsymbol{\xi}_t; \boldsymbol{S}_{\iota t} \mid \mathcal{I}_{\iota,t-1}) \right\}, \tag{10}$$

[8] The forecasts of other agents enter the target through these loadings.

[9] We focus on the steady-state solution of the problem, in which the uncertainty about the state, conditional on past information, has converged, so that the attention strategy and the Kalman gain are time-invariant; a single pair, rather than a sequence of pairs indexed by time, then describes the strategy.

subject to the law of motion Equation (7), the signal equation Equation (9), the mapping Equation (8) from $\boldsymbol{\xi}_t$ to the target $\boldsymbol{x}_t^*$, and the optimality condition $\boldsymbol{x}_{\iota t} = E[\boldsymbol{x}_t^* \mid \mathcal{I}_{\iota t}]$. The matrix $\boldsymbol{\Theta}$, negative definite, weighs the losses incurred from the deviation of the actions from their optimal values; when the problem is microfounded, $\boldsymbol{\Theta}$ is the Hessian of the agent's objective with respect to its actions, evaluated at the non-stochastic steady state. The scalar $\lambda > 0$ is the marginal cost of attention, and $I(\boldsymbol{\xi}_t; \boldsymbol{S}_{\iota t} \mid \mathcal{I}_{\iota,t-1})$ is the mutual information between the state and the signal conditional on past information, which measures the per-period reduction in uncertainty.

Afrouzi and Yang (2021) characterize the solution: the optimal signals load on the directions of the state in which the marginal benefit of information—the gain from tracking the target more closely today plus the value of a more precise prior tomorrow—exceeds its cost, and their number never exceeds the number of actions. The attention strategy therefore depends on the equilibrium only through the mapping from the state to the target, $\boldsymbol{\Psi}^*$. The problem can be posed with attention paid to all shocks jointly, or to each shock as a distinct activity. On the history of innovations, the state has $n_\varepsilon T$ coordinates, which makes the standard iteration on the first-order condition costly; Appendix A.1 describes how we solve the problem at that scale.

3.3. Mapping Signals to Expectation Matrices. Given the signals, the estimate $\hat{\boldsymbol{\xi}}_{\iota t} := E[\boldsymbol{\xi}_t \mid \mathcal{I}_{\iota t}]$ follows from the Kalman filter:

$$\hat{\boldsymbol{\xi}}_{\iota t} = \boldsymbol{F}\hat{\boldsymbol{\xi}}_{\iota,t-1} + \boldsymbol{K}\,\boldsymbol{\varsigma}_{\iota t}, \qquad \boldsymbol{\varsigma}_{\iota t} := \boldsymbol{S}_{\iota t} - \boldsymbol{\Gamma}\boldsymbol{F}\hat{\boldsymbol{\xi}}_{\iota,t-1}, \tag{11}$$

with $\boldsymbol{\varsigma}_{\iota t}$ the surprise in the signal and $\boldsymbol{K}$ the steady-state gain, whose expression and Riccati recursion are given in Appendix A.2. Since the state is the history of innovations, the filtered estimate contains the estimate of every past innovation, and the filter delivers its revisions over time. Averaging Equation (11) across the group, the noise integrates to zero, and the average estimate $\bar{\boldsymbol{\xi}}_t$ follows:

$$\bar{\boldsymbol{\xi}}_t = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{\Gamma})\boldsymbol{F}\,\bar{\boldsymbol{\xi}}_{t-1} + \boldsymbol{K}\boldsymbol{\Gamma}\,\boldsymbol{\xi}_t. \tag{12}$$

The expectation matrix is obtained by feeding an innovation at date 0 into Equation (12) and iterating it for T periods, starting from $\bar{\boldsymbol{\xi}}_{-1} = \boldsymbol{0}$. The average estimate obtained at date h gives row h of the expectation matrix: its entry s is the average estimate, at date h, of the innovation at date $h - s$.[10] Given the signals, the expectation matrix does not depend on the target, and applies to whatever target the inner loop produces.

[10] With several shocks, each entry becomes a block of n_ε estimates, and the recursion is run once for each innovation.

The history of innovations is the most transparent state and, in practice, the most accurate one, but not the cheapest. When the targets admit an exact low-order VARMA representation, the attention problem is computationally easier to solve on its state-space realization, which has a handful of coordinates instead of $n_\varepsilon T$. Appendix A.3 describes this compact state and how the estimates of past innovations are recovered on it.

4. Algorithm

4.1. The Fixed Point. Given the expectation matrices $\boldsymbol{\mathcal{E}} := (\boldsymbol{\mathcal{E}}_g)_g$, the equilibrium is the linear system of the inner loop:

$$\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}})\,\boldsymbol{x} + \boldsymbol{\Phi}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0}, \tag{13}$$

where $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}})$ is $\boldsymbol{\Phi}_x$ with each group's optimality conditions replaced by Equation (6). The expectation matrices, however, depend on the equilibrium, $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}(\boldsymbol{x})$, so that an equilibrium is a fixed point:

$$\boldsymbol{\Phi}_x\big(\boldsymbol{\mathcal{E}}(\boldsymbol{x})\big)\,\boldsymbol{x} + \boldsymbol{\Phi}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0}. \tag{14}$$

Two maps are involved. Given the sequences, the map $\boldsymbol{x} \mapsto \boldsymbol{\mathcal{E}}(\boldsymbol{x})$ solves each group's attention problem and builds its expectation matrix from the chosen signals (Section 3); this map is non-linear—the inherent difficulty of rational inattention, under which the signals themselves adjust to the equilibrium. Given the expectation matrices, the map $\boldsymbol{\mathcal{E}} \mapsto \boldsymbol{x}$ is a linear solve. The fixed point can be stated in either object, and the choice of the object iterated on leaves the equilibrium unchanged.

4.2. Iterating on Expectation Matrices. Our algorithm iterates on the expectation matrices and solves the equilibrium exactly in between:

1. Start from perfect information, $\boldsymbol{\mathcal{E}}^{(0)} = \boldsymbol{I}$, and solve $\boldsymbol{\Phi}_x(\boldsymbol{I})\boldsymbol{x}^{(0)} + \boldsymbol{\Phi}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0}$: the perfect information equilibrium.
2. Given $\boldsymbol{x}^{(n)}$, solve each group's attention problem and build its expectation matrix: $\tilde{\boldsymbol{\mathcal{E}}}^{(n+1)} = \boldsymbol{\mathcal{E}}(\boldsymbol{x}^{(n)})$.
3. Damp: $\boldsymbol{\mathcal{E}}^{(n+1)} = (1-\omega)\boldsymbol{\mathcal{E}}^{(n)} + \omega\tilde{\boldsymbol{\mathcal{E}}}^{(n+1)}$, with $\omega > 0$.[11]

[11] The admissible range of ω depends on the strength of the feedback of the paths on the expectation matrices; values $\omega > 1$ over-relax the update.

4. Solve $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}}^{(n+1)})\boldsymbol{x}^{(n+1)} + \boldsymbol{\Phi}_\varepsilon\boldsymbol{\varepsilon} = \mathbf{0}$, and return to step 2 until convergence.[12]

The part of $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}})$ that does not involve the expectation matrices is built once; only the rows of the groups' conditions change across iterations.

4.3. Iterating on Aggregate Paths. Existing procedures instead iterate on the aggregate paths that agents take as given. In our notation, they run as follows:

1. Guess the dynamics of the aggregate variables entering the targets, $\boldsymbol{x}^{(0)}$.
2. Given $\boldsymbol{x}^{(n)}$, solve each group's attention problem and build its expectation matrix: $\boldsymbol{\mathcal{E}}(\boldsymbol{x}^{(n)})$.
3. Set the average actions of each group to $\boldsymbol{\mathcal{E}}_g(\boldsymbol{x}^{(n)})\,\boldsymbol{x}^*_g(\boldsymbol{x}^{(n)})$, with the targets evaluated at the guess, and obtain the remaining variables from the other equilibrium conditions given those actions: $\tilde{\boldsymbol{x}}^{(n+1)}$.
4. Damp: $\boldsymbol{x}^{(n+1)} = (1-\omega)\boldsymbol{x}^{(n)} + \omega\tilde{\boldsymbol{x}}^{(n+1)}$, and return to step 2 until convergence.

Step 2 is common to both procedures; the difference lies in step 3. Because the targets are evaluated at the guess, actions and targets are never solved jointly: the equilibrium feedback—from the price level to the firms' target and back to the price level, say—is worked out across iterations, through the damping of the guess. In our algorithm, the same feedback is resolved within each iteration, and only the expectation matrices are damped.

Where the feedback is resolved matters for convergence. In the *tâtonnement*, the damping must contract the feedback of the equilibrium on itself, which is often strong—strategic complementarity in price setting, the response of the wage to labour demand—so the step has to be small and convergence takes many iterations. In our algorithm, the damping only has to contract the feedback of the paths on the expectation matrices, which is weak by comparison: a change in the paths moves the agents' estimates of the innovations much less than it moves their targets. Larger steps, and far fewer iterations, therefore suffice. Nor is an iteration costlier: obtaining the implied aggregates in step 3 already requires solving the rest of the economy given the actions, whereas our step 4 is one linear solve whose structure is fixed.

4.4. Sensitivity to the Initial Guess. Step 1 of our algorithm starts from perfect information, $\boldsymbol{\mathcal{E}}^{(0)} = \boldsymbol{I}$, but nothing in the iteration requires it. We check that the solution does not depend on this starting point by solving the small-scale model

[12] Convergence is declared when two successive iterates, whether of the expectation matrices, the implied actions or the aggregate paths, differ by less than a tolerance.

of Maćkowiak and Wiederholt (2015) from 32 other initial expectation matrices, ranging from zero attention to the solutions of neighboring calibrations and the expectations implied by randomly drawn targets.[13] From every starting point, the algorithm converges to the same expectation matrices, up to the tolerance, in 19 to 32 iterations, against 21 from perfect information. In practice, the choice of the starting point can thus be used to save iterations: in estimation, each parameter draw can start from the expectation matrices of a nearby draw. In the models of Table 1, starting from the solution at information costs 5% higher cuts the number of iterations by two thirds to 85%. Whether the solution depends on the starting point should nonetheless be checked in each application, since strategic complementarities in information acquisition are known to generate multiple equilibria in some settings (Hellwig and Veldkamp, 2009; Yang, 2015).

5. Applications

5.1. Solving Models from the Literature. We solve the models of Afrouzi and Yang (2021), Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), described in Appendix B, with their published procedures and with ours, and holding fixed the calibration, the state of the attention problems used in the original paper.[14]

At the authors' step, our procedure needs fewer iterations in every model, by a factor of 2.5 in Afrouzi and Yang (2021) and of 12 in Maćkowiak and Wiederholt (2015). In Maćkowiak and Wiederholt (2025), whose steps of 2×10^{-4} and 5×10^{-4} require tens of thousands of iterations under either procedure, this brings no gain in time. The main gain comes from the step size: because our iteration solves the equilibrium exactly, it remains stable at steps of 0.2 to 0.5, at which it converges in 30 to 80 iterations and under 40 seconds in Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), 28 to more than 1,500 times faster than the published procedures. In Afrouzi and Yang (2021), the authors' step of 1 is the fastest of those we tried for our procedure, and the gain is a factor of 2.6.[15] All procedures reach the same solution: the information flows agree to the third decimal.

5.2. Likelihood-Based Estimation. *(TBC)*

[13] One with zero attention, 12 taken from the solutions of neighboring calibrations, and 19 implied by random ARMA(2,1) targets.

[14] When the attention problems are solved on the state-space realization of a (V)ARMA representation of the targets, the expectation matrices are recovered as described in Appendix A.3.

[15] For Afrouzi and Yang (2021), our procedure also admits over-relaxation, i.e., $\omega > 1$. We do not report the result here.

Table 2: Iterations and Solution Time

			Published		Ours, same step		Ours, step 0.5	
Model	Shock	Step	Iter.	Time	Iter.	Time	Iter.	Time
AY21	Monetary & technology	1	508	219	206	85.0	414	163
MW15	Monetary	0.1	2,014	925	171	22.0	34	5.1
	TFP	0.1	1,901	125	163	21.0	32	4.5
MW25	Technology, $h = 0$	$2{\cdot}10^{-4}$	174,873	14,743	117,052	17,939	35	9.6
	Technology, $h = 2$	$5{\cdot}10^{-4}$	35,737	3,674	35,047	4,379	81	39.8

Notes: Time in seconds. In MW25, the technology shock is anticipated h periods ahead ($h = 0$: no news). In MW15 and MW25, where firms and households are both inattentive, we solve in two stages, first with inattentive firms only, then with both, and sum the iterations over the two. For MW25 with $h = 2$, the largest admissible step is below 0.5; we use 0.2.

6. Conclusion

We propose a new algorithm for solving RI-DSGE models that decouples attention from the general equilibrium. Given attention, an inner loop solves exactly for the paths of all endogenous variables in the sequence space; the outer loop then updates attention given those paths. Because general equilibrium feedbacks are resolved within each iteration rather than across iterations, the outer loop tolerates large steps and converges in few iterations, which in turn makes it affordable to solve the attention problems on an almost exact representation of the equilibrium dynamics. On the models of Afrouzi and Yang (2021), Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), our algorithm reaches the same equilibrium as the published procedures 2.6 to 1,536 times faster.

With the companion toolbox, RI-DSGE models with several groups of inattentive agents, rich dynamics and long horizons can be solved in seconds where the published procedures take up to hours, fast enough for likelihood-based estimation. In Briand (2026), we apply our method to a medium-scale RI-DSGE model and estimate it using Bayesian methods.

From a broader perspective, our method extends the reach of expectation matrices in the sequence space. Auclert, Rognlie, and Straub (2020) use them with exogenous information frictions; we take them to economies in which agents learn from endogenous variables, so that the forecasts of others matter and the expectation matrices themselves depend on the equilibrium. Rational inattention is one instance: the algorithm applies whenever agents filter Gaussian signals about the state, whether these signals are specified exogenously, as in Section 3.1, or follow

from another model of information choice that, given the equilibrium paths, delivers such signals.

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A. Attention: Technical Details

A.1. Solving LQG-RI on Large States With Gaussian signals, the mutual information in Equation (10) equals $\frac{1}{2}\left[\log\det\boldsymbol{\Sigma}_{t|t-1} - \log\det\boldsymbol{\Sigma}_{t|t}\right]$, where $\boldsymbol{\Sigma}_{t|t-1}$ and $\boldsymbol{\Sigma}_{t|t}$ are the prior and posterior covariances of the state (Appendix A.2). In steady state, the posterior solves a first-order condition in which the marginal benefit of information is the gain from tracking the target more closely today—a quadratic form in the state, given by $\boldsymbol{\Theta}$ and the mapping Equation (8)—plus the discounted value of a more precise prior tomorrow (Afrouzi and Yang, 2021). Writing $\boldsymbol{\Omega}$ for this benefit, the optimal signals are on the eigenvectors of:

$$\boldsymbol{\Sigma}_{t|t-1}^{1/2}\,\boldsymbol{\Omega}\,\boldsymbol{\Sigma}_{t|t-1}^{1/2} \tag{15}$$

whose eigenvalues exceed $\lambda/2$, with noise chosen so that the corresponding eigenvalues of $\boldsymbol{\Sigma}_{t|t}^{1/2}\boldsymbol{\Omega}\boldsymbol{\Sigma}_{t|t}^{1/2}$ equal $\lambda/2$; there are at most as many such signals as actions, n_a (their Theorem 2.2).

We solve this first-order condition with the fixed-point map of Afrouzi and Yang (2021), which iterates on the prior covariance and $\boldsymbol{\Omega}$. On the history of innovations, the state has $n := n_\varepsilon T$ coordinates, and each step of the map requires two eigendecompositions of a dense $n \times n$ matrix: one for the square root of the prior, one for Equation (15). We avoid both. First, any square root of the prior will do: with its Cholesky factor $\boldsymbol{C}$, the matrix $\boldsymbol{C}'\boldsymbol{\Omega}\boldsymbol{C}$ has the same eigenvalues as Equation (15), and eigenvectors rotated by an orthogonal matrix that cancels in every quantity of interest. Second, only the attended eigenpairs matter: we find the top $n_a + 1$ by subspace iteration on $\boldsymbol{v} \mapsto \boldsymbol{C}'\boldsymbol{\Omega}(\boldsymbol{C}\boldsymbol{v})$, without forming the matrix and starting from the subspace found at the previous step of the map; the (n_a+1)-th eigenvalue falling below $\lambda/2$ certifies that no attended direction is missed. A step then costs one Cholesky factorization, $O(n^3)$, and a subspace iteration, $O(n_a n^2)$, and the map converges to the same fixed point. Table 3 compares the two on three problems of Maćkowiak, Matějka, and Wiederholt (2018) and on one with two targets.[16]

When the attention problems are solved many times, as in our outer loop or in estimation, we recommend warm-starting each one from the prior covariance, $\boldsymbol{\Omega}$ and attended subspace of the previous solution, which is then close to its fixed point.

A.2. The Filter The gain of the Kalman filter Equation (11) is

$$\boldsymbol{K}_t = \boldsymbol{\Sigma}_{t|t-1}\boldsymbol{\Gamma}'\left(\boldsymbol{\Gamma}\boldsymbol{\Sigma}_{t|t-1}\boldsymbol{\Gamma}' + \boldsymbol{\Sigma}_{\boldsymbol{\nu}}\right)^{-1}, \tag{16}$$

[16] The four problems are included as examples in our toolbox.

Table 3: Attention Problems: Solution Time

Problem	State	n	Time (seconds) AY21	Ours	Speed-up
AR(2)	MA(100)	101	62.5	3.28	19
	MA(200)	201	225	8.04	28
	MA(400)	401	990	13.7	72
Price level	MA(100)	101	7.59	0.142	53
	MA(200)	201	48.9	0.393	124
	MA(400)	401	307	1.58	194
Two targets	MA(100)	202	10.6	0.232	46
	MA(200)	402	31.0	0.917	34
	MA(400)	802	134	3.86	35
News	MA(100)	101	2.50	0.403	6.2
	MA(200)	201	7.42	1.33	5.6
	MA(400)	401	31.4	4.59	6.8

Notes: Each row is one attention problem with given targets: three examples of Maćkowiak, Matějka, and Wiederholt (2018)—an AR(2) target, price setting (the state has a unit root), and a target that responds to a shock announced 8 periods ahead—and a problem with two targets and two shocks. MA(T) is the history of the last T innovations; n is the dimension of the state. AY21 is the fixed-point map of Afrouzi and Yang (2021); both maps run at the same step, tolerance and iteration cap, and agree to within 10^{-6}. On the minimal ARMA state, both solve every problem in a few milliseconds. Numerical optimization over parameterized signals, as in Maćkowiak and Wiederholt (2015), is in general both slower on minimal states and inaccurate on large ones, with errors in the information flow of up to 1.1 bits. Hardware: AMD EPYC 7532, 4 cores; Julia 1.10.

where the prior covariance $\boldsymbol{\Sigma}_{t|t-1} := \mathrm{Var}(\boldsymbol{\xi}_t \mid \mathcal{I}_{\iota,t-1})$ and the posterior covariance $\boldsymbol{\Sigma}_{t|t} := \mathrm{Var}(\boldsymbol{\xi}_t \mid \mathcal{I}_{\iota t})$ evolve according to the Riccati recursion

$$\boldsymbol{\Sigma}_{t|t} = \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{K}_t \boldsymbol{\Gamma} \boldsymbol{\Sigma}_{t|t-1}, \qquad \boldsymbol{\Sigma}_{t+1|t} = \boldsymbol{F} \boldsymbol{\Sigma}_{t|t} \boldsymbol{F}' + \mathrm{diag}\left(\boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}}, \mathbf{0}, \ldots, \mathbf{0}\right). \tag{17}$$

In steady state, the covariances and the gain are at the fixed point of Equation (17), and we drop the time subscripts, with $\boldsymbol{\Sigma}$ the prior covariance.

A.3. Compact States When the variables the signals load on admit an exact low-order VARMA representation, the filter can be run on its state-space realization, $\boldsymbol{\zeta}_{t+1} = \boldsymbol{A}\boldsymbol{\zeta}_t + \boldsymbol{B}\boldsymbol{\varepsilon}_{t+1}$, which has a handful of coordinates instead of $n_\varepsilon T$. That state does not contain the past innovations, whose estimates follow instead from their

projection on the surprises observed since they occurred:

$$E[\boldsymbol{\varepsilon}_\tau \mid \mathcal{I}_{\iota t}] = \sum_{d=\tau}^{t} \boldsymbol{P}_{d-\tau}\, \boldsymbol{\varsigma}_{\iota d}, \qquad \boldsymbol{P}_\ell = \boldsymbol{\Sigma}_{\boldsymbol{\varepsilon}} \boldsymbol{B}' \left[(\boldsymbol{A}(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{\Gamma}))' \right]^\ell \boldsymbol{\Gamma}' \boldsymbol{\Sigma}_{\boldsymbol{\varsigma}}^{-1}, \tag{18}$$

with $\boldsymbol{\Sigma}_{\boldsymbol{\varsigma}} := \boldsymbol{\Gamma}\boldsymbol{\Sigma}\boldsymbol{\Gamma}' + \boldsymbol{\Sigma}_{\boldsymbol{\nu}}$ the variance of the surprise. The orders of the VARMA are unrestricted and may change from one iteration to the next. When no low-order VARMA fits the sequences—a target still moving at the horizon, a news shock with a long delay—the history of innovations is used, and it is in any case the reference against which the compact state is checked.[17]

B. Models

All variables are in log-deviations from the non-stochastic steady state. Rationally inattentive agents set their action to the conditional expectation of their target, with the same losses as in the original papers. Table 4 reports the calibrations.

B.1. Afrouzi-Yang 2021. Firms set prices under rational inattention, and households have perfect information. The target of firms is:

$$p_t^* = p_t + \tfrac{\gamma+\psi}{1+\psi\theta}\, x_t,$$

and the price level p_t is the average of the firms' estimates of this target. The remaining equilibrium conditions are:

$$\begin{aligned}
\pi_t &= p_t - p_{t-1}, \\
x_t &= E_t x_{t+1} - \gamma^{-1}\left(i_t - E_t \pi_{t+1}\right) + E_t y_{t+1}^n - y_t^n, \\
y_t^n &= \tfrac{1+\psi}{\gamma+\psi}\, a_t, \\
y_t &= x_t + y_t^n, \\
i_t &= \rho_r\, i_{t-1} + (1-\rho_r)\left(\phi_\pi \pi_t + \phi_x x_t + \phi_{\Delta y} \Delta y_t\right) + u_t, \\
a_t &= \rho_a a_{t-1} + \sigma_a \varepsilon_t^a, \\
u_t &= \sigma_u \varepsilon_t^u,
\end{aligned}$$

with x_t the output gap and y_t^n natural output.

[17] Time-domain least squares on the impulse responses; the lowest orders within a tolerance, kept across iterations while the fit stays within it; unit roots fitted in first differences. The fit is refused, not degraded, when it misses a sequence by more than a tenth of a percent of its sum of squares.

B.2. Maćkowiak-Wiederholt 2015. Firms set prices, and households set consumption and their wage, under rational inattention; prices and wages are otherwise flexible. The targets of firms, for prices, and of households, for consumption and the wage, are:

$$\begin{aligned} p_t^* &= p_t + \tfrac{mc_t}{1+\theta(1-\varphi)/\varphi}, \\ c_t^* &= c_{t+1}^* - \gamma^{-1}\left(i_t - \pi_{t+1}\right), \\ w_t^* &= \tfrac{\gamma c_t^* + \psi n_t + \psi\theta_w w_t}{1+\theta_w\psi}, \end{aligned}$$

and the price level p_t, consumption c_t and the real wage w_t are the averages of the agents' estimates of these targets. The remaining equilibrium conditions are:

$$\begin{aligned} y_t &= a_t + \varphi n_t, \\ mc_t &= w_t + (1-\varphi)n_t - a_t, \\ \pi_t &= p_t - p_{t-1}, \\ i_t &= \rho_r i_{t-1} + (1-\rho_r)\left(\phi_\pi \pi_t + \phi_x(y_t - y_t^n)\right) + u_t, \\ y_t^n &= \tfrac{1+\psi}{\varphi\gamma+\psi+1-\varphi}\, a_t, \\ y_t &= c_t, \\ a_t &= \rho_a a_{t-1} + \sigma_a \varepsilon_t^a, \\ u_t &= \rho_u u_{t-1} + \sigma_u \varepsilon_t^u, \end{aligned}$$

with n_t hours worked and mc_t the real marginal cost.

B.3. Maćkowiak-Wiederholt 2025. Firms choose capital and labor, and households choose consumption and labor supply, under rational inattention; the households' budget constraint ties consumption to labor income and dividends, so their attention problem bears on labor supply alone. The targets of firms, for capital and labor demand, and of households, for labor supply, are:

$$\begin{aligned} k_t^* &= \tfrac{E_t a_{t+1} - \varphi w_{t+1} - \frac{1-\varphi}{1-\beta(1-\delta)}\, r_t}{1-\alpha-\varphi}, \\ l_t^{d*} &= \tfrac{a_t - w_t}{1-\varphi}, \\ l_t^{s*} &= w_t - \gamma\left(\omega_W w_t + \omega_D d_t\right), \end{aligned}$$

and capital k_t, the composite $l_t - \frac{\alpha}{1-\varphi}k_{t-1}$ and scaled labor supply $(\gamma\omega_W + \psi)\, l_t$ are the averages of the agents' estimates of these targets. The remaining equilibrium

conditions are:

$$\begin{aligned}
y_t &= a_t + \alpha k_{t-1} + \varphi l_t, \\
inv_t &= \tfrac{k_t - (1-\delta)k_{t-1}}{\delta}, \\
d_t &= \tfrac{y_t - \varphi(w_t + l_t) - I_Y\, inv_t}{D_Y}, \\
c_t &= \omega_W (w_t + l_t) + \omega_D d_t, \\
r_t &= \gamma \left(c_{t+1} - c_t\right), \\
a_t &= \rho_a a_{t-1} + \sigma_a \varepsilon^a_{t-h},
\end{aligned}$$

with inv_t investment, d_t dividends, $I_Y = \alpha\beta\delta/(1 - \beta(1 - \delta))$, $D_Y = 1 - \varphi - I_Y$, $\omega_W = \varphi/(1 - I_Y)$ and $\omega_D = D_Y/(1 - I_Y)$. The technology innovation is announced h periods in advance; $h = 0$ is the standard technology shock and $h > 0$ a news shock. As in Maćkowiak and Wiederholt (2025), the benefit of information is not discounted in the attention problems.

Table 4: Calibrations

				MW25	
		AY21	MW15	$h=0$	$h=2$
β	Discount factor	0.99	0.99	0.99	0.99
γ	Inverse IES	2.5	1	1	1
ψ	Inverse Frisch elasticity	2.5	0	10^{-5}	10^{-5}
θ	Elasticity of demand	10	4	–	–
θ_w	Elasticity of substitution between labor types	–	4	–	–
φ	Labor elasticity	–	2/3	0.65	0.65
α	Capital elasticity	–	–	0.33	0.33
δ	Depreciation rate	–	–	0.025	0.025
λ_f	Firms' cost of attention	1.546	2.89×10^{-4}	6×10^{-5}	6.2×10^{-5}
λ_h	Households' cost of attention	–	2.89×10^{-4}	1×10^{-5}	3×10^{-5}
ϕ_π	Taylor rule, inflation	2.028	1.5	–	–
ϕ_x	Taylor rule, output gap	0.168	0.125	–	–
$\phi_{\Delta y}$	Taylor rule, output growth	3.122	–	–	–
ρ_r	Interest rate smoothing	0.9457	0.85	–	–
ρ_a	Persistence of technology	0.85	0.93	0.9	0.9
σ_a	Std. of technology innovations	1.56	0.64	0.008	0.008
ρ_u	Persistence of monetary shocks	0	0	–	–
σ_u	Std. of monetary innovations	0.279	0.119	–	–
T	Truncation horizon	200	250	400	400

Notes: Costs of attention are per nat in AY21 and per bit in MW15 and MW25. In MW25, h is the news horizon.